

TECHSTORESys

Detailed System Documentation

Information Technology Directorate
Zimbabwe National Army - Tech Stores & Procurement

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Comprehensive reference: architecture, roles, modules, procurement workflow, stakeholder portals, creditors, data model, API, and appendices. Visual companions: [app/system-flow.html](#) and [app/data-flow-diagram.html](#).

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1. Executive Summary

TECHSTORESys is the IT Directorate integrated stores, requisitions, procurement, and accountability system for the Zimbabwe National Army (ZNA). It digitises Tech Stores ledgers (GL accounts), issue and receipt vouchers (ZNA Q 1033), unit requisitions, the DP procurement cycle, stakeholder portals (GS Branch, DAF, DP, AIAD, Supplier), the Creditors register, workshop repairs, ICT equipment accountability, and management reporting.

The application is a browser-based single-page system (SPA) served locally by server.py. Application state persists in SQLite via REST /api/state. A portable Windows build (PyInstaller) ships in dist/TECHSTORES-Portable for deployment without a separate Python install.

1.1 Design principles

- Role-scoped navigation - users see only modules permitted for their role.
- Single source of truth - one appState synced to server; modules share data.
- Procurement traceability - every buy tracked from requisition through payment.
- ZNA form fidelity - digital Q forms mirror official store paperwork.
- Offline resilience - localStorage fallback when server unavailable.

2. System Architecture

2.1 Components

- Frontend SPA: app/index.html, app/css/main.css, app/js/*.js
- Module loader: lazy-loads app/modules/{moduleId}.html fragments.
- Backend: server.py - ThreadingHTTPServer, SQLite, JSON API.
- Boot chain: boot.js -> auth.js -> config.js -> module scripts -> dashboard.js.
- Portable EXE: TECHSTORES.exe wraps server; TECHSTORES-LAUNCHER opens browser.

2.2 Deployment modes

Mode	Launcher	Description
Full system	START-SYSTEM.bat	Python server + default browser on localhost
Portable	OPEN-TECHSTORES.bat	Bundled EXE + app folder on USB/desktop
Offline shell	START-OFFLINE.bat	Precached assets; limited API

2.3 Directory layout

Path	Purpose
app/	UI, modules, CSS, JS, flow diagrams
app/js/	Business logic per functional area
app/modules/	HTML fragments loaded on demand
data/	SQLite DB, exchange rate, product cache
assets/	Icons, nav images, ZNA crest
scripts/	Build, PDF, parse utilities
dist/	Shipped portable executables

3. Users, Roles and Access Control

Login uses username/password against appState.users. Each user has a role key mapped in ROLE_PERMISSIONS (app/js/config.js). Permissions: canEdit, canReleaseCut, canManageUsers, canBackup, canReports. Oversight roles (Director, DD, AQSO2, RQ, Army Commander, Brigadiers) receive modules=["*"] with canEdit=false.

3.1 Stakeholder desk roles

Role key	Portal window	Module bundle
gs_sd	GS Branch	MODULES_DESK_GS
dir_daf	DAF	MODULES_DESK_DAF + Creditors
dir_dp	DP Contracts	MODULES_DESK_DP
dir_aiad	Due Diligence	MODULES_DESK_AIAD
supplier	Supplier	MODULES_DESK_SUPPLIER

3.2 Default demo accounts (change in production)

server.py ships DEFAULT_PLAIN_PASSWORDS for demo: admin/admin123, rq/rq123, daf/daf123, dp/dp123, gsdesk/gsdesk123, aiad/aiad123, orderly/orderly123, etc. Replace before operational use.

4. Module Catalog (by function)

4.1 Command & oversight

- dashboard - KPIs, alerts watch, role-scoped home shortcuts.
- universal-search - cross-module search with history.
- reports-module - TechStores period and template reports.
- process-guides - Learning Centre with procurement charts.
- system-help / system-dictionary - terminology reference.

4.2 Requisitions & correspondence

- unit-requisitions - in-tray with age, stock status, intelligence panel.
- orderly-room - Daily File mirror; correspondence routing.
- doc-import - AI/server parse; apply extracted fields to forms.
- it-dir-comms - department messaging portal.

4.3 Stores & GL ledgers

- Five GL ledgers: office supplies, software, maintenance, spare parts, ICT equipment.
- voucher-module (ZNA Q 1033) - issues/receipts with ZA numbers.
- delivery-note, stock-take, unit-checks, accommodation-stores.
- financial-year-bids, release-cut (admin), gl-targets monthly proposals.
- temporary-loans (14-day) and permanent-loans (Comd/34).

4.4 ICT accountability

- ict-accountability - ZNA ICT Asset Register; typeable unit picker.
- ict-distribution, unit-equipment, monthly-returns.
- techstores-equipment-register, gate-register (RP).
- workshop-repairs, workshop-receipt-cert.

4.5 Procurement

- dp-f1-form, cost-comparative-schedule, spec-evaluation.
- dp-procurement - 8-step ICT procurement cycle tracker.
- portals-board - visual workflow dashboard with deep links.
- stakeholder-desk - GS/DAF/DP/AIAD/Supplier windows.
- purchase-orders, suppliers-contracts, undelivered-orders, supplier-debts (Creditors).

4.6 Workshop buy tools

- laptop-compare - duty-profile ranking from local catalog.
- ict-compare - web crawl H2H compare; buy-score chart.
- guide-quotation - rough estimate helper.

4.7 ZNA Q / SVCS forms

22+ digital forms including Q 1033, Q 982, Q 178, Q 1043, Q 80, SVCS 890, Q 1179, Q 987, Q 3977, Q 985, Q 1, Q 998, Q 1680, Q 3, Q 31, Q 40, Q 1049, Q 1229, Q 1571, Q 1954, SVCS 1045. Index: zna-q-forms-index.

5. End-to-End Operational Flows

5.1 Requisition to issue (in stock)

Unit submits requisition -> Orderly Room / Requisitions in-tray -> RQ checks stock on GL ledger -> if available, raise ZNA Q 1033 issue voucher -> record ZA number on ICT Accountability if applicable -> close requisition.

5.2 Requisition to purchase (out of stock)

Requisition marked out-of-stock -> raise DP F1 -> GS endorsement -> MANAC (DAF) -> PFMS number -> DP call for quotes -> IT Dir spec eval -> AIAD due diligence -> P/O -> supply/D-note -> delivery verified -> Creditors entry -> DAF payment.

5.3 Creditors lifecycle

Goods received and verified -> Creditors case opened -> chased_daf when memo sent -> DAF pays -> upload paid-list or mark paid -> status paid. Undelivered Items covers the opposite case (ordered, not yet received).

6. ICT Procurement Workflow

Step	Status code	Responsible actor
1	spec_raise_f1	IT Dir / Unit
2b	awaiting_gs	Colonel SD (GS Branch)
2c	awaiting_manac	DAF
2d	pfms_req_no	IT Dir RQ
3	f1_with_dp	DP Contracts
3b	quotes_itdir_eval	IT Dir / Workshop
3c	spec_returned_dp	DP
4	aiad_due_diligence	AIAD
4b	aiad_cert_issued	AIAD
5	po_raised / po_manual_pending_daf	DP / DAF
6	supply_delivery	Supplier
7	delivery_verified	IT Dir
8	payment_complete	DAF

7. Stakeholder Portals

Each portal (stakeholder-desk.js) shows the full in-tray for that actor from their entry point onward, not only active-step cases. Cases needing action sort first and display an Action marker.

7.1 GS Branch

- Queue: all cases from awaiting_gs onward plus GS stakeholder touch.
- Action: awaiting_gs without gs.endorsedAt - upload signed endorsement.

7.2 DAF

- Tab Procurement cycle: MANAC, payment vouchers, procurement pay queue.
- Tab Creditors & payment: stats strip, intelligence, Excel import, paid-list.
- Action: awaiting_manac, po_manual_pending_daf, delivery verified unpaid.

7.3 DP

- Queue: from f1_with_dp onward plus quotes_itdir_eval and DP touch.
- Action: call quotes, record winner, send AIAD, raise P/O.

7.4 AIAD

- Queue: from aiad_due_diligence / spec_returned_dp onward.
- Action: issue Price Due Diligence certificate; upload signed form.

7.5 Supplier

- Cases visible when supplier named on P/O or invited in stakeholder slot.
- Upload quotation, spec compliance, D-note, invoice, banking details.

8. Creditors Register

- Module id: supplier-debts (nav label: Creditors).
- Seed: it-dir-creditors-seed.js - IT DIR register 05 Nov 2025.
- Import: POST /api/creditors/parse - drag-drop Excel in module or DAF tab.
- Paid list: POST /api/creditors/parse-paid - batch mark matched suppliers paid.
- Intelligence: priority chase, duplicate supplier groups, merge helper.

- Statuses: open, chased_daf, part_paid, paid, cancelled.
- Chase memo printable for DAF action.

9. Smart / Intelligence Features

- Requisitions: respond-first (urgent + oldest), overdue/out-of-stock flags.
- Undelivered: priority supplier follow-up ranking.
- Creditors: DAF chase queue, duplicate detection, paid-list matching.
- Alerts desk: cross-module watch with navigateToModule deep links.
- ICT compare: ictCompareHistory avoids repeat web crawls.
- Typeable select (utils.js): combobox with type-to-filter; used for ZNA units.

10. Data Persistence and API

10.1 Storage modes

storageMode: online | offline-shell | offline-local | local-only. Auto-save debounced to PUT /api/state. SQLite file under data/.

10.2 REST API

Endpoint	Method	Purpose
/api/health	GET	Health check
/api/state	GET/PUT	Load/save appState
/api/login	POST	Authenticate user
/api/audit	POST	Audit log entry
/api/mode	GET	Current storage mode
/api/mode/switch	POST	Switch online/offline
/api/exchange-rate	GET	RBZ USD/ZWL cache
/api/product-specs	POST	Product spec enrichment
/api/product-enrich	POST	Product web enrich
/api/product-images	POST	Product image fetch
/api/market-catalog	GET/POST	Market catalog cache
/api/creditors/parse	POST	Parse creditors Excel
/api/creditors/parse-paid	POST	Parse DAF paid-list Excel
/api/ai/ask	POST	Offline assistant
/api/ai/import-document	POST	Document import parse
/api/ai/spec-document	POST	Spec document parse
/api/ai/draft-justification	POST	Draft justification text

11. User Interface Conventions

- Sidebar nav grouped: Stores, Requisitions, Procurement, Workshop, etc.
- Role filter hides unauthorized nav items at render time.
- navigateToModule(id, opts) - opts: stkDesk, stkDafTab, sdId for deep links.
- ZNA unit picker: wireZnaUnitPicker / fillZnaUnitSelect - establishment groups.
- Print: body.printing-* classes hide chrome; voucher and memo layouts.
- Cache bust: ?v=YYYYMMDDtag on script/link tags in index.html.

12. Operations and Maintenance

- Start: START-SYSTEM.bat or python server.py (default port from server.py).
- Build portable: scripts/build-portable.bat (icon: assets/techstores.ico).
- Regenerate icon: python scripts/build-icon.py.
- Backup: admin menu; copy data/*.db for cold backup.
- Regenerate this PDF: python scripts/generate-system-documentation-pdf.py.

12.1 Related files

- app/system-flow.html + TECHSTORESys-System-Flow.pdf
- app/data-flow-diagram.html + TECHSTORESys-Data-Flow-Diagram.pdf
- TECHSTORESys-Progress-Mon31Aug-Tue1Sep-2026.pdf - recent changes

13. Glossary

DAF

Directorate of Army Finance - MANAC endorsement and supplier payment

DP F1

Directorate Procurement Form 1 - authority to procure

MANAC

Management Accounting - DAF funds endorsement on F1

PFMS

Public Financial Management System - ministry requisition number

AIAD

Army Internal Audit Directorate - price due diligence audit

GS Branch

General Staff - Colonel SD endorses F1 before MANAC

Q 1033

ZNA Issue and Receipt Voucher for store movements

ZA No.

ZNA asset number assigned to issued ICT equipment

Creditors

Suppliers who delivered goods/services but are not yet paid

Undelivered

Orders placed with supplier but goods not yet received

Release Cut

Admin transfer of buying power between GL votes

H2H Compare

Head-to-head ICT equipment comparison via web crawl

Typeable select

Combobox control: type to filter long option lists

Stakeholder desk

Role-specific portal window for external directorates

Window

Full-screen module opened from nav or dashboard (form-container)

Form

Fillable data-entry screen inside a module window

14. Application Windows

In TECHSTORESys a window is a full-screen module opened from the sidebar, dashboard shortcut, or portals board. Each window loads an HTML fragment from app/modules/ and is controlled by matching JavaScript. Windows are role-filtered at login.

14.1 Role-scoped home workspaces

On login, non-RQ roles see a tailored dashboard home (dept-home.js) with shortcuts to their primary windows.

Role preset	Window title	Primary screens
gate / rp	RP Gate workspace	Gate Register - ICT in/out at Directorate gate
storeman	Storeman workspace	Q 1033, delivery notes, stock take, loans, ZNA Q forms
workshop	Workshop workspace	Workshop Register, spec eval, laptop/H2H compare
orderly	Orderly Room workspace	Daily File, correspondence, doc import
comms	Department comms	IT Dir Comms portal only
dp	DP Window	Stakeholder desk + P/O, suppliers, procurement cycle
gs	GS Branch Window	Stakeholder desk + F1 endorsement, requisitions
daf	DAF Window	Stakeholder desk + Creditors tab, financial year bids
aiad	Due Diligence Window	Stakeholder desk + spec / cost comparative
supplier	Supplier Window	Stakeholder desk - quotes, D-note, banking

14.2 Stakeholder portal windows

All five external directorate windows share module stakeholder-desk. The desk parameter (data-stk-desk) selects GS, DAF, DP, AIAD, or Supplier chrome. Layout: case list (left) + case detail (right). DAF has a second tab for Creditors.

Desk key	Window name	Purpose
dp	DP Window	Quotations, adjudication, P/O, AIAD handoff; full DP in-tray
gs	GS Branch Window	F1 endorsement upload; cases from awaiting_gs onward
daf	DAF Window	Tab 1: MANAC/payment. Tab 2: Creditors import and chase
aiad	Due Diligence Window	Price due diligence certificate; AIAD pipeline
supplier	Supplier Window	RFQ/P/O response; upload quote, spec, invoice

- Open via: sidebar Portals submenu, portals-board tiles, or role home shortcuts.
- Deep link: `navigateToModule('stakeholder-desk', { stkDesk: 'daf', stkDafTab: 'creditors' })`.

14.3 IT Directorate department desks

Eight internal department windows (dept-desk.js). Each has: demand/requisition form, inbox, demands log, and Establishment panel (nav sub-link). Submit copies Dir, DD, AQSO2 and TechStores.

Module ID	Desk title	Function
dept-sysadmin	Systems Administration Desk	Demand form CC Dir/DD/AQSO2/TechStores
dept-workshop	Workshop Desk	Workshop spares/tools demand; link to Workshop Register
dept-compengr	Computer Engineering / DBA Desk	Servers, storage, network engineering items
dept-swengr	Software Engineering Desk	Dev tools, licences, project support
dept-ictsec	ICT Security Desk	Security tools, certificates, appliances
dept-itts	ITTS Desk	Training school equipment and media
dept-admin	Admin Office Desk	Admin office ICT and internal requirements
dept-gate	Gate / RP Desk	Gate equipment demand; link to Gate Register

14.4 Other major windows

Module ID	Window	Notes
portals-board	Portals dashboard	Clickable procurement workflow map
dp-procurement	ICT Procurement Cycle	8-step tracker for all DP cases
it-dir-comms	IT Dir Comms	Department messaging portal
doc-import	Import Document	Parse scan/text into form fields
dashboard	Dashboard Overview	KPIs, alerts, universal search
gl-3112210001	GL ICT Equipment ledger	One of five GL account windows
laptop-compare	Laptop Compare	Catalog duty-profile ranking
ict-compare	H2H ICT Comparison	Web crawl compare with buy-score chart

14.5 Window UI pattern

- Header bar: form-title + close button (returns to dashboard).
- class form-container on root div; id matches module ID for navigation.
- Print: body.printing-{moduleId} hides other containers.
- Module loader caches loaded HTML; init*Module() runs on first open.

15. Forms Catalogue

Forms are fillable data-entry screens inside module windows. ZNA Q forms follow Annex A Sec 1 Ch 7 (ASO). Procurement forms follow DP/DAF/AIAD cycle. Registers are structured lists with add/edit/print rather than single-page official layouts.

15.1 ZNA Q and SVCS forms (official stores forms)

22 forms are implemented as fillable modules. The full ASO catalogue (90+ entries) is indexed at [zna-q-forms-index](#); reference-only forms are listed for completeness but open on paper.

Form	Title	Module ID
Q 1	Statement of stores lost or damaged to be written off	zna-q-1
Q 3	Issue to a Government department on repayment	zna-q-3
Q 31	Cash purchase/receipt	zna-q-31
Q 40	Artisan tools list	zna-q-40
Q 80	Schedule of stores boarded / Ledger sheet	zna-q-80
Q 178	Ledger (sub / distribution ledger sheet)	zna-q-178
Q 982	Combined indent and voucher for stores	zna-q-982
Q 985	Discrepancy report	zna-q-985
Q 987	Certificate of stocktaking	zna-q-987
Q 998	Statement of loss, damage or destruction	zna-q-998
Q 1033	Issue and receipt voucher (large and small)	zna-q-1033
Q 1043	Report on stores and condemnation certificate	zna-q-1043
Q 1049	Transfer voucher	zna-q-1049
Q 1157	Clothing and personal equipment record	zna-q-1157
Q 1179	Clothing and necessities issue voucher	zna-q-1179
Q 1229	Certificate of accidental breakage	zna-q-1229
Q 1571	Debit voucher	zna-q-1571
Q 1680	Miscellaneous credit/debit voucher	zna-q-1680
Q 1954	Recoveries from individuals	zna-q-1954
Q 3977	Equipment/vehicle neglect, misuse and damage report	zna-q-3977
SVCS 890	Demand / Issue	zna-svcs-890
SVCS 1045	Workshop indent	zna-svcs-1045

- Primary daily form: Q 1033 (also voucher-module shortcut in nav).
- Q 982 combined indent links to procurement indent workflow.
- SVCS 1045 used for workshop stores indents.

15.2 Procurement and evaluation forms

Module ID	Form name	Use in cycle
dp-f1-form	DP F1 ITDIR Form	Official indent for purchasing/procurement
cost-comparative-schedule	Cost Comparative Schedule	Vendor price comparison grid
spec-evaluation	Spec/Tech Evaluation	Technical evaluation of supplier quotes
guide-quotation	Rough Guide Quotation	Quick estimate before formal quotes
purchase-orders	Purchase Orders	P/O register - electronic and manual/DAF
delivery-note	Delivery Note	Supplier delivery documentation
workshop-receipt-cert	Workshop Receipt Certificate	Workshop receipt of repaired items

Module ID	Form name	Use in cycle
financial-year-bids	Financial Year Bids	Annual bid submissions and votes

15.3 Operational registers (structured forms)

Module ID	Register	Purpose
unit-requisitions	Requisitions in-tray	Incoming unit/formation requests
orderly-room	Orderly Room Daily File	Correspondence and requisition mirror
supplier-debts	Creditors register	Non-paid goods received; DAF chase
undelivered-orders	Undelivered Items	Ordered but not yet received
ict-accountability	ZNA ICT Asset Register	Equipment accountability by unit
voucher-module	Issue/Receipt Voucher	Alias screen for ZNA Q 1033
temporary-loans	Temporary Loans	14-day controlled stores loans
permanent-loans	Permanent Loans	Laptops/iPads under Comd/34
workshop-repairs	Workshop Register	Repairs intake and status
gate-register	Gate Register	RP gate movements in/out
stock-take	Stock Take	Full stores inventory count
monthly-returns	Monthly Returns	Unit ICT equipment monthly returns

15.4 Form features

- Doc Import: upload or paste document; server parses and fills target form fields.
- ZNA unit picker: typeable combobox on unit/formation fields (zna-units.js).
- Print layouts: most Q forms and DP F1 have print-optimised CSS.
- Table search: many forms include data-search-target toolbar filters.
- Auto-save: edits persist to appState on change (debounced server sync).

Appendix A: Module ID Reference

Complete list of 76 registered module IDs and display labels.

Module ID	Display label
dashboard	Dashboard Overview
orderly-room	Orderly Room - DF & Correspondence Files
it-dir-comms	IT Directorate - Communications Portal
dept-sysadmin	Dept Desk - Systems Administration
dept-workshop	Dept Desk - Workshop
dept-compengr	Dept Desk - Computer Engineering
dept-swengr	Dept Desk - Software Engineering
dept-ictsec	Dept Desk - ICT Security
dept-itts	Dept Desk - ITTS
dept-admin	Dept Desk - Admin Office
dept-gate	Dept Desk - Gate / RP
gl-2200600002	GL 6122100009 - ZOFF / Office Supplies & Services
gl-2200600003	GL 2200600003 - Software Licenses
gl-220200002	GL 220200002 - Tech Equipment Maintenance
gl-2201900002	GL 2201900002 - Spare Parts
gl-3112210001	GL 3112210001 - ICT Equipment
voucher-module	Issue Voucher / ZNA-Q-1033
stock-take	Stock Take - Full Stores Inventory
unit-checks	Unit Check Log - ASO Ch 28
financial-year-bids	Financial Year Bids
unit-equipment	Unit Equipment
ict-accountability	ZNA ICT Asset Register
ict-distribution	ICT Equipment Distribution Lists
temporary-loans	Temporary Loans - Controlled Stores
permanent-loans	Permanent Loans - Laptops & iPads
monthly-returns	Monthly Returns - Unit ICT Equipment
undelivered-orders	Undelivered Items
supplier-debts	Creditors - non-paid goods / DAF payment register
workshop-receipt-cert	Workshop Receipt Certificate
delivery-note	Delivery Note
purchase-orders	Purchase Orders
accommodation-stores	Inventory of Accommodation Stores
unit-requisitions	Requisitions - In-tray
doc-import	Import Document
spec-evaluation	Spec/Tech Evaluation
laptop-compare	Laptop Compare - buy the winner
ict-compare	H2H ICT Comparison - crawl & compare
guide-quotation	Rough Guide Quotation

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Module ID	Display label
dp-f1-form	DP F1 Form
cost-comparative-schedule	Cost Comparative Schedule
dp-procurement	ICT Procurement Cycle
portals-board	Portals - procurement workflow dashboard
stakeholder-desk	Portals - DP / GS / DAF / AIAD / Supplier
workshop-repairs	Workshop Register
gate-register	Gate Register (RP)
techstores-equipment-register	TechStores Equipment Register
suppliers-contracts	Suppliers and Contracts
zna-q-forms-index	SUMMARY OF Q FORMS - Index
zna-q-982	ZNA Q 982 - Combined Indent
zna-q-178	ZNA Q 178 - Sub Ledger Sheet
zna-q-1033	ZNA Q 1033 - Issue & Receipt Voucher
zna-q-1043	ZNA Q 1043 - Condemnation Certificate
zna-q-80	ZNA Q 80 - Ledger Sheet
zna-svcs-890	ZNA SVCS/890 - Demand / Issue
zna-q-1179	ZNA Q 1179 - Clothing Issue Voucher
zna-q-987	ZNA Q 987 - Stocktaking Certificate
zna-q-3977	ZNA Q 3977 - Neglect / Damage Report
zna-q-1157	ZNA Q 1157 - Clothing & Equipment Record
zna-q-985	ZNA Q 985 - Discrepancy Report
zna-q-1	ZNA Q 1 - Stores lost/damaged write-off
zna-q-998	ZNA Q 998 - Loss / Damage / Destruction
zna-q-1680	ZNA Q 1680 - Miscellaneous credit/debit voucher
zna-q-3	ZNA Q 3 - Issue on repayment
zna-q-31	ZNA Q 31 - Cash purchase / receipt
zna-q-40	ZNA Q 40 - Artisan tools list
zna-q-1049	ZNA Q 1049 - Transfer voucher
zna-q-1229	ZNA Q 1229 - Accidental breakage certificate
zna-q-1571	ZNA Q 1571 - Debit voucher
zna-q-1954	ZNA Q 1954 - Recoveries from individuals
zna-svcs-1045	ZNA SVCS 1045 - Workshop Indent
duties-roles	Duties & Roles
process-guides	Learning Centre - Process & Charts
system-help	System Help / Dictionary
reports-module	Reports
user-management	User Management
release-cut	Release Cut

Appendix B: Role Permission Matrix

Summary of ROLE_PERMISSIONS flags. Modules column describes scope; (*) = all modules.

Role	Modules	Edit	Release cut	Users	Reports
admin	All (*)	Yes	Yes	Yes	Yes
director / dd / aqso2 / rq / tso	All (*)	No (view)	No	Yes	Yes
army_commander / brig_*	All (*)	No (view)	No	Yes	Yes
store_officer / rq	Stores + procurement	Yes	No	No	Yes
storeman	Issues, loans, ZNA Q	Yes	No	No	No
workshop / oc_workshop	Workshop + compare	Yes	No	No	No
orderly_clerk / oc_admin	Orderly + comms	Yes	No	No	No
gs_sd	GS portal + F1	Yes	No	No	No
dir_daf	DAF portal + creditors	Yes	No	No	Yes
dir_dp	DP portal + P/O	Yes	No	No	Yes
dir_aiad	AIAD portal + spec	Yes	No	No	Yes
supplier	Supplier portal	Yes	No	No	No
rp / oc_gate	Gate register	Yes	No	No	No
oc_* comms desks	IT Dir comms only	Yes	No	No	No
viewer	Dashboard + help	No	No	No	Yes

Desk module lists (config.js)

- MODULES_DESK_GS: portals + requisitions + dp-f1 + dp-procurement
- MODULES_DESK_DAF: portals + dp-procurement + supplier-debts + financial-year-bids
- MODULES_DESK_DP: portals + dp-procurement + spec + P/O + undelivered + suppliers
- MODULES_DESK_AIAD: portals + dp-procurement + cost-comparative + spec-evaluation
- MODULES_DESK_SUPPLIER: portals shared bundle only
- MODULES_RQ: full stores ledgers + procurement + reports (no orderly-room)

Appendix C: Application State Keys

Top-level keys persisted in appState / SQLite (server.py STATE_KEYS subset):

State key
users
auditLog
glLedgers
glMonthlyTargets
storesInventory
customInventoryLedgers
customCatalogItems
stockTakes
monthlyReturns
ictAccountability
ictDistributionLists
requisitions

State key
orderlyDailyFile
correspondenceFiles
undeliveredOrders
supplierDebts
workshopReceiptCerts
dpProcurements
costComparativeSchedules
unitChecks
officeMessages
alertDesk
ictCompareHistory
navMenuOrder

Appendix D: ZNA Q Forms (fillable in system)

Complete list of official ZNA Q / SVCS forms with digital modules. Source: app/js/zna-q-catalogue.js. Index module: zna-q-forms-index.

Form no.	Official title	Module ID	Nav path
Q 1	Statement of stores lost or damaged to be written off	zna-q-1	Sidebar > ZNA Q Forms > Q 1
Q 3	Issue to a Government department on repayment	zna-q-3	Sidebar > ZNA Q Forms > Q 3
Q 31	Cash purchase/receipt	zna-q-31	Sidebar > ZNA Q Forms > Q 31
Q 40	Artisan tools list	zna-q-40	Sidebar > ZNA Q Forms > Q 40
Q 80	Schedule of stores boarded / Ledger sheet	zna-q-80	Sidebar > ZNA Q Forms > Q 80
Q 178	Ledger (sub / distribution ledger sheet)	zna-q-178	Sidebar > ZNA Q Forms > Q 178
Q 982	Combined indent and voucher for stores	zna-q-982	Sidebar > ZNA Q Forms > Q 982
Q 985	Discrepancy report	zna-q-985	Sidebar > ZNA Q Forms > Q 985
Q 987	Certificate of stocktaking	zna-q-987	Sidebar > ZNA Q Forms > Q 987
Q 998	Statement of loss, damage or destruction	zna-q-998	Sidebar > ZNA Q Forms > Q 998
Q 1033	Issue and receipt voucher (large and small)	zna-q-1033	Sidebar > ZNA Q Forms > Q 1033
Q 1043	Report on stores and condemnation certificate	zna-q-1043	Sidebar > ZNA Q Forms > Q 1043
Q 1049	Transfer voucher	zna-q-1049	Sidebar > ZNA Q Forms > Q 1049
Q 1157	Clothing and personal equipment record	zna-q-1157	Sidebar > ZNA Q Forms > Q 1157
Q 1179	Clothing and necessities issue voucher	zna-q-1179	Sidebar > ZNA Q Forms > Q 1179
Q 1229	Certificate of accidental breakage	zna-q-1229	Sidebar > ZNA Q Forms > Q 1229
Q 1571	Debit voucher	zna-q-1571	Sidebar > ZNA Q Forms > Q 1571
Q 1680	Miscellaneous credit/debit voucher	zna-q-1680	Sidebar > ZNA Q Forms > Q 1680
Q 1954	Recoveries from individuals	zna-q-1954	Sidebar > ZNA Q Forms > Q 1954
Q 3977	Equipment/vehicle neglect, misuse and damage report	zna-q-3977	Sidebar > ZNA Q Forms > Q 3977
SVCS 890	Demand / Issue	zna-svcs-890	Sidebar > ZNA Q Forms > SVCS 890
SVCS 1045	Workshop indent	zna-svcs-1045	Sidebar > ZNA Q Forms > SVCS 1045

Appendix E: Procurement and register forms

Category	Module ID	Form / register name
Procurement	dp-f1-form	DP F1 ITDIR Form
Procurement	cost-comparative-schedule	Cost Comparative Schedule
Procurement	spec-evaluation	Spec/Tech Evaluation
Procurement	guide-quotation	Rough Guide Quotation
Procurement	purchase-orders	Purchase Orders
Procurement	delivery-note	Delivery Note
Procurement	workshop-receipt-cert	Workshop Receipt Certificate
Procurement	financial-year-bids	Financial Year Bids
Register	unit-requisitions	Requisitions in-tray
Register	orderly-room	Orderly Room Daily File
Register	supplier-debts	Creditors register
Register	undelivered-orders	Undelivered Items

Category	Module ID	Form / register name
Register	ict-accountability	ZNA ICT Asset Register
Register	voucher-module	Issue/Receipt Voucher
Register	temporary-loans	Temporary Loans
Register	permanent-loans	Permanent Loans
Register	workshop-repairs	Workshop Register
Register	gate-register	Gate Register
Register	stock-take	Stock Take
Register	monthly-returns	Monthly Returns

Document control

Auto-generated v1.2 on 01 September 2026 from TECHSTORESys source. Authoritative module/role definitions: app/js/config.js. For visual architecture see app/system-flow.html and app/data-flow-diagram.html.